

Chakradhar Peela

Chennai, Tamil Nadu, India | peelachakradhar@gmail.com | linkedin.com/in/chakradhar-peela-231b64218 | github.com/entusedglue3125

PROFESSIONAL SUMMARY

Passionate Electronics & Communication Engineering student at Saveetha School of Engineering (SIMATS) specializing in AI/ML, IoT, Robotics, Cybersecurity, and Next-Gen Networks (5G/6G). Experienced in building AI agents, robotic systems, and intelligent embedded devices that bridge hardware and software. GDG On Campus Lead at SIMATS with a proven track record of organizing tech events, building communities, and rapidly prototyping over 250+ engineering projects.

TECHNICAL SKILLS

- **Programming:** C / C++, Python, AI/ML, SQL, HTML, CSS
- **Embedded Systems & Electronics:** ESP32, Arduino, Raspberry Pi, STM32, PCB Design, Sensors & Actuators
- **Software Tools:** MATLAB/Simulink, Blender / FreeCAD, ModelSim, KiCad
- **Networking & IoT:** 5G/6G Protocols, MQTT / CoAP, WiFi / BLE, LoRa / Zigbee
- **Core Interests:** AI/ML & Deep Learning, Computer Vision, Cybersecurity, Signal Processing, ROS, Robotics, Gazebo

EDUCATION

B.E. Electronics & Communication Engineering 2024 - Present

Saveetha School of Engineering (SIMATS), Chennai, Tamil Nadu

EXPERIENCE & HARDWARE PROJECTS

AI-Integrated Robotics & Embedded Systems Developer 2025 - Present

Independent & Hackathon Projects

- Built the first AI-linked Robotic Arm running fully on-device utilizing a Raspberry Pi ecosystem.
- Developed expertise in ROS1, ROS2, Gazebo, AI Automation pipelines, and sophisticated hardware control mechanisms.
- Completed and deployed over 270+ projects involving Robotics and Embedded Systems, scaling up rapid prototyping massively.
- Mastered AI integration methodologies with Large Language Models and advanced workflow automation using n8n for 5G/6G projects.
- Formed a competitive hackathon team and actively won engineering competitions.

Key Engineering Projects

NeuralArm

- Engineered a neural network-controlled robotic arm using Python, MediaPipe, OpenCV, and Raspberry Pi (PCA9685).
- Achieved real-time gesture and pose tracking where the hardware arm mirrors human hand/body movements.
- Implemented real-time data transmission via UDP protocols from a laptop processing unit to the Raspberry Pi controller.

A.R.E.S (Autonomous Robotic & Embedded System)

- Developed a comprehensive dashboard and control interface for autonomous embedded systems.
- Created HTML/JavaScript-based real-time monitoring and control panels for efficient system management and observability.

Face Recognition Attendance System

- Built an automated attendance system using Python and OpenCV that detects and recognizes faces in real-time via webcam.
- Engineered continuous comparison against a pre-stored database to log attendance data into CSV datasets automatically.

Cybersecurity & AI Workflows

- Developed a collection of AI agents, automation workflows, and cybersecurity experiments.
- Implemented N8N orchestration, gesture-controlled Jarvis AI agents, and local Ollama deployments to automate complex tasks.

LEADERSHIP & ACTIVITIES

GDG On Campus Lead

Saveetha School of Engineering (SIMATS)

- Organized technical events, workshops, and built a community around cutting-edge technologies.

Member

Youth United Council of India

- Participated in social impact initiatives and youth campaigns.

AWARDS & HONORS

- **Best Paper Award - ICRIMR 2025** (International Conference on Recent Innovations in Multidisciplinary Research) - Presented innovative technology research, earning top honors.